

Temperature- Monitoring/Warning System

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Project aim and requirements

Project Aim

This project aimed to create an embedded temperature-monitoring/warning system with the use of an Arduino Mega 2560. The system must have a constant reading of the ambient temperature using the DHT11 sensor and a constant display of the temperature on a 16x2 LCD. When the temperature exceeds the assigned threshold, a warning LED is activated along with a warning message on the LCD. The system should also be able to detect a failed sensor reading, triggering a fault condition. The final stage would be designing a custom Arduino Mega shield PCB to replace the breadboard's messy wiring.

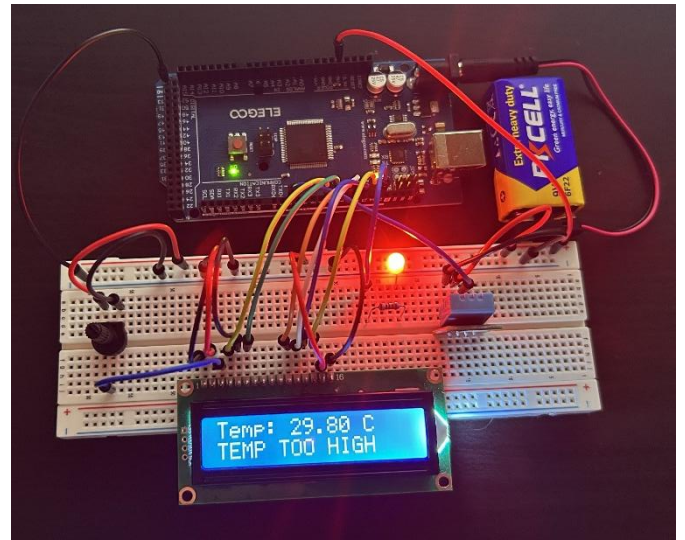
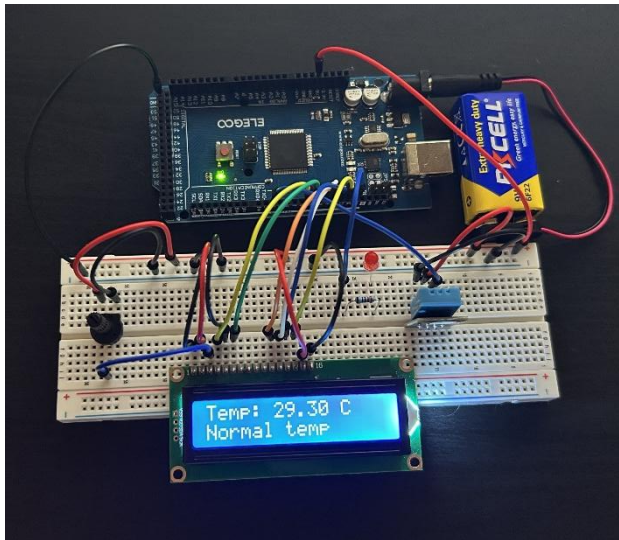
Requirements

- The system measures the temperature using a DHT11 sensor
- The measured temperature is displayed on a 16x2 LCD
- The system must compare the measured temperature to the warning threshold
- When the temperature is at or above the threshold, the warning LED must turn on
- When the temperature is at or above the threshold, the LCD must display a high temperature warning message
- Under normal temperature conditions, the LED will be off, and the LCD will read "normal temperature"
- An invalid DHT11 sensor reading should be detected, a sensor fault message displayed, and the LED activated
- The system must operate independently after programming with a battery power source
- A custom PCB should be designed for an Arduino Mega shield to replace the breadboard wiring.

Hardware used

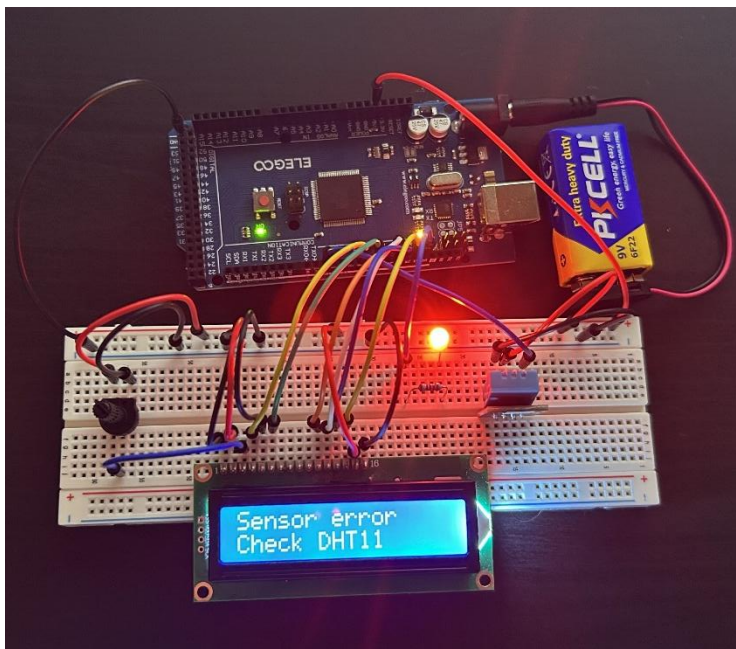
- ELEGOO Arduino Mega 2560
- DHT11 temperature and humidity module
- 1602A V2.0 16x2 LCD
- 10 k Ω potentiometer
- LED
- 220 Ω resistor
- Breadboard
- Jumper wires
- Battery
- Custom Arduino Mega shield PCB designed in KiCad

Breadboard prototype



The temperature threshold was set to 29.8°C to trigger the temperature warning, as the DHT11 was reading the room temperature.

To test the DHT11 sensor error, I unplugged the sensor



Software/control logic

sketch_aug12b.ino

```
1  #include <DHT.h>           // DHT sensor library
2  #include <LiquidCrystal.h> // 16x2 LCD library
3
4  #define DHTPIN 2           // DHT11 data pin
5  #define DHTTYPE DHT11     // sensor type
6  #define LEDPIN 13          // warning LED output pin
7
8  LiquidCrystal lcd(7, 8, 9, 10, 11, 12); // RS, E, D4, D5, D6, D7, LCD pin allocation
9
10 DHT dht(DHTPIN, DHTTYPE); // create DHT11 sensor
11
12 void setup() {
13     dht.begin(); // initialise the DHT11 sensor
14
15     Serial.begin(9600); // serial monitor for testing
16
17     pinMode(LEDPIN, OUTPUT); // sets warning LED pin as an output
18
19     lcd.begin(16, 2); // initialise the 16-column, 2-row LCD
20 }
21
22 void loop() {
23     float temperature = dht.readTemperature(); // read temperature in degrees Celsius
24
25     if (isnan(temperature)) { // if the value given by the DHT11 is an invalid reading
26         Serial.println("ERROR: Failed to read from DHT11");
27
28         lcd.setCursor(0, 0);
29         lcd.print("Sensor error      ");
30
31         lcd.setCursor(0, 1);
32         lcd.print("Check DHT11      ");
33
34         digitalWrite(LEDPIN, HIGH); // LED on for faulty DHT11 reading
35         delay(2000);
36         return;
37     }
38
39     Serial.println(temperature);
40
41     lcd.setCursor(0, 0);
42     lcd.print("Temp: ");
43     lcd.print(temperature);
44     lcd.print(" C");
45
46     if (temperature >= 29.8){ // temporary threshold for testing
47         lcd.setCursor(0, 1);
48         lcd.print("TEMP TOO HIGH");
49         Serial.println("WARNING TEMP TOO HIGH");
50         digitalWrite(LEDPIN, HIGH);
51     }
52     else {
53         lcd.setCursor(0, 1);
54         lcd.print("Normal temp ");
55         digitalWrite(LEDPIN, LOW);
56     }
57     delay(1000); // wait 1 second before taking the next sensor reading
58 }
```

Testing

Test	Expected result	Result
Temperature below threshold	LED off and LCD says normal temperature	Pass
Temperature above threshold	LED on and warning message on LCD	Pass
Sensor disconnected	Error message on LCD and LED on	pass
Battery powered	System runs independently	pass

The DHT11 sensor was checked using **isnan(temperature)** to identify a failed reading. If the reading was invalid, the LED turns on, and the LCD displays a sensor error message.

To simulate a sensor fault, I temporarily disconnected the DHT11 signal connection.

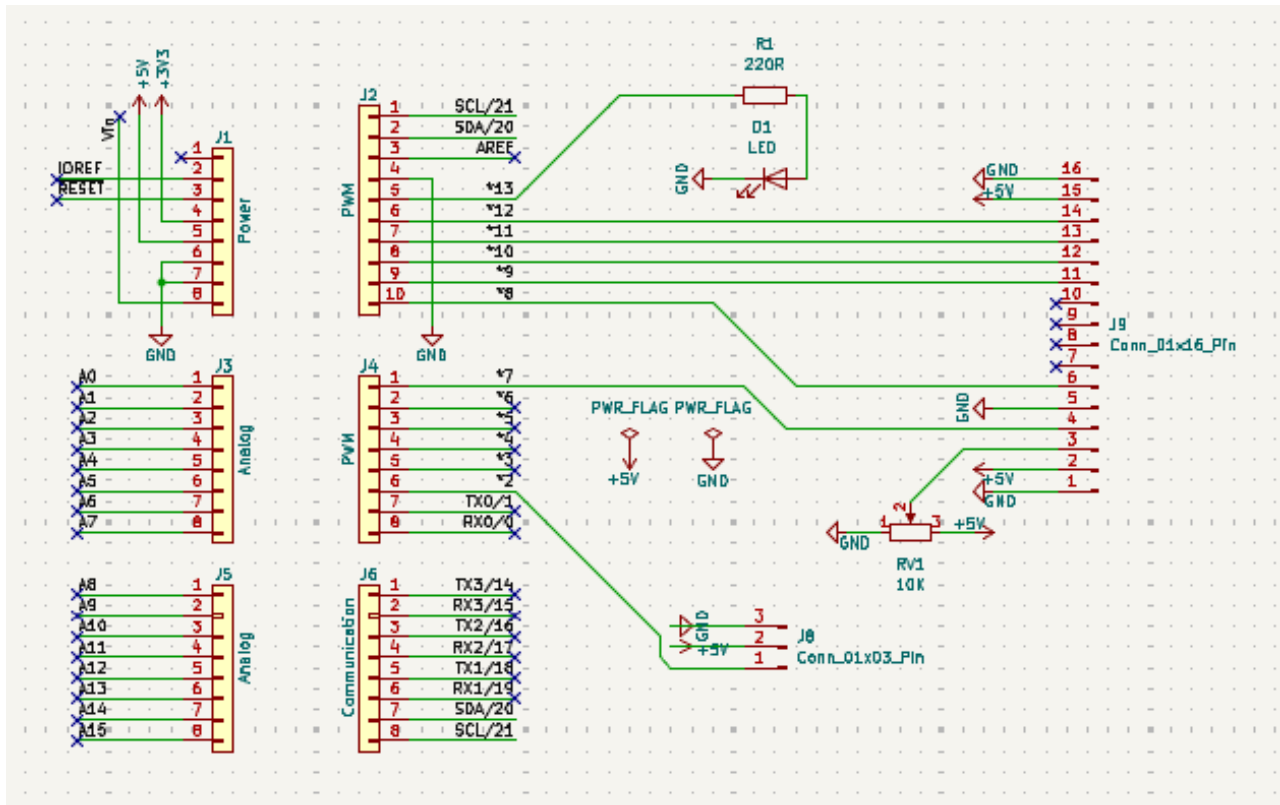
Problems encountered

Problem	Cause	Fix
LED not lighting up	Both LED legs were accidentally placed on the same breadboard path	Moved the legs to separate rows
LCD not displaying properly	D7 was incorrectly connected to backlight pin A	Moved the wire for D7 to the correct data pin
A failed DHT11 reading would leave the system reading invalid data	A broken or improper connection of the data pin or the power supply	Adding isnan(temperature) to check if the data is valid or not, resulting in an error message displayed and the warning LED on
PCB ERC/DRC issues	The unused Arduino Mega template pins and SDA/SCL connections created warnings/errors	Used no connect flags where appropriate; added PWR_FLAG for supply to +5V/GND, and excluded understood template-caused violations

PCB design

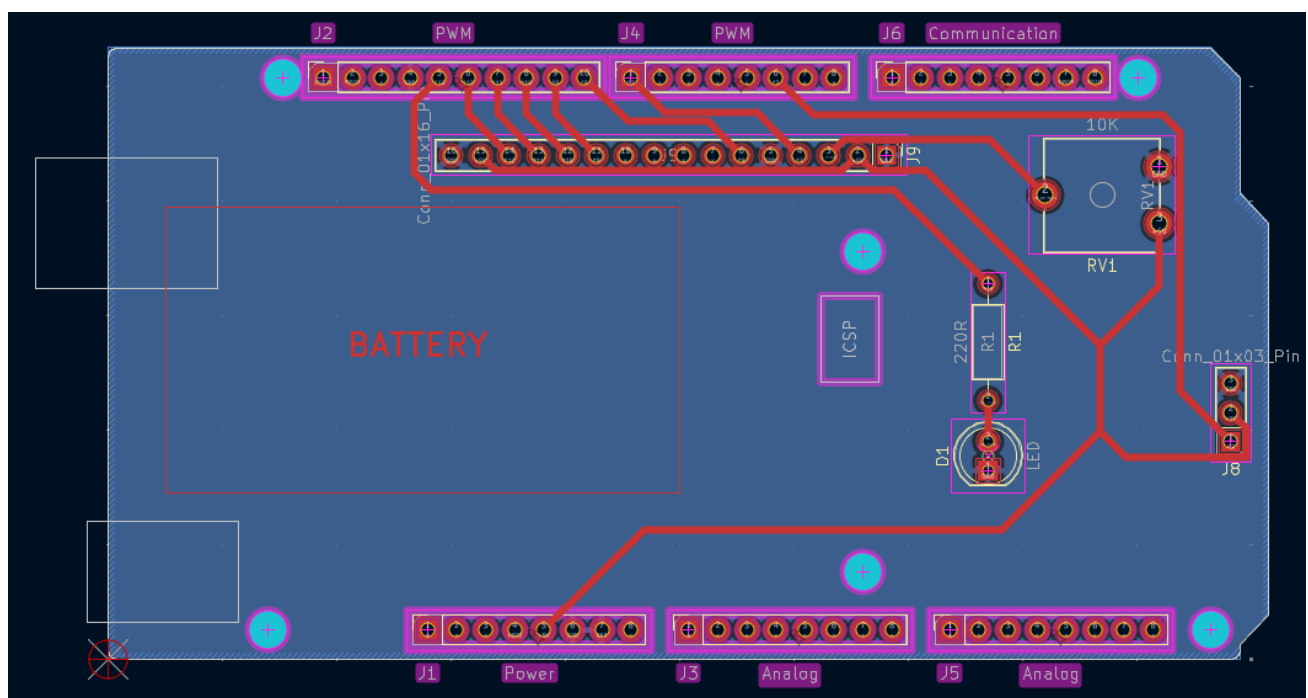
KiCad schematic:

The ERC was used to confirm the schematic was correct

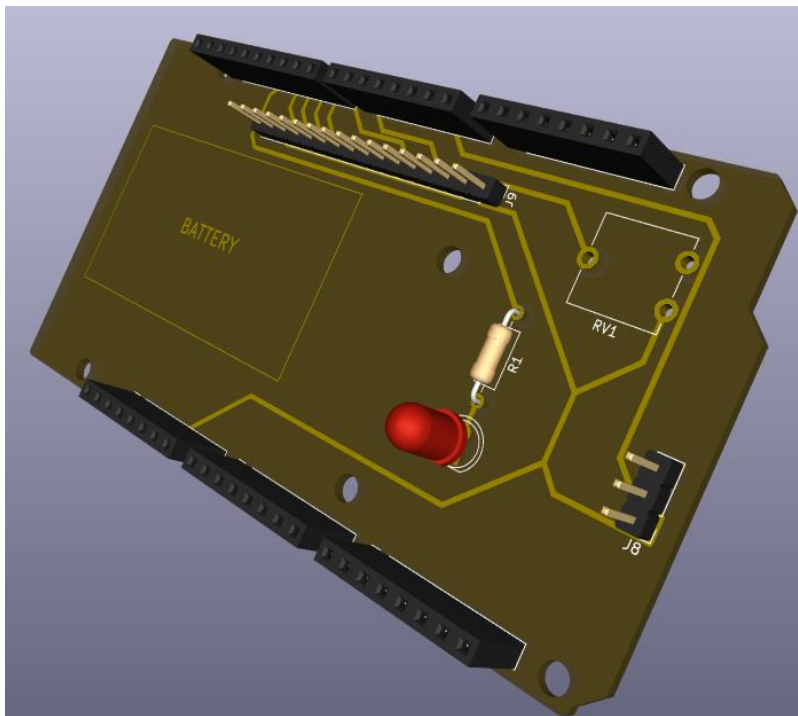


KiCad PCB layout:

The DRC was used to confirm the design was correct.



3D view of the PCB shield:



I designed the PCB as an Arduino Mega shield so it can plug directly onto the Mega 2560. All female headers will be replaced with male headers on the bottom of the PCB to allow it to plug into the Mega 2560. And the male headers on top will be changed to female headers, so the 16x2 LCD and DHT11 can be connected and remain removable.

The footprints assigned for the PCB design can be changed as needed when soldering on the headers. All that is required is the correct spacing and connections.

I measured my components in real life to choose the correct footprints needed.

Final result

The temperature was successfully measured using the DHT11 and displayed on the 16x2 LCD. The warning LED worked for both high temperature and the sensor fault; the fault-detection logic for the sensor also worked as required, with the LCD displaying a warning/error message. The final breadboard design could operate independently from the laptop with a battery once programmed; finally, the breadboard prototype was made into a custom Arduino Mega shield PCB design.

The final physical stage is to manufacture and solder the custom PCB. This will be completed when soldering equipment is available at my university.

Possible future improvements

A temperature-controlled fan could be added that uses PWM (Pulse Width Modulation) based on the temperature, allowing the cooling rate to increase or decrease when needed.

To ensure the system never needs a temperature threshold code change again, some user-adjustable buttons to alter the threshold could be added.

The manufacturing and soldering of the custom PCB need to be completed.

Eventually, the Arduino Mega can be replaced with a microcontroller mounted directly on the PCB.

References

Arduino (n.d.) *Mega 2560 Rev3*. Available at: <https://docs.arduino.cc/hardware/mega-2560/> (Accessed: 17 August 2026).

Arduino (n.d.) *Liquid Crystal Displays (LCD) with Arduino*. Available at: <https://docs.arduino.cc/learn/electronics/lcd-displays> (Accessed: 17 August 2026).

Adafruit Industries (n.d.) *DHT sensor library*. Available at: <https://github.com/adafruit/DHT-sensor-library> (Accessed: 17 August 2026).

KiCad Developers Team (2025) *Getting Started in KiCad 9.0*. Available at: https://docs.kicad.org/9.0/en/getting_started_in_kicad/getting_started_in_kicad.html (Accessed: 17 August 2026).

KiCad Developers Team (n.d.) *PCB Editor 9.0*. Available at: <https://docs.kicad.org/9.0/en/pcbnew/pcbnew.html> (Accessed: 17 August 2026).

Aosong Electronics Co., Ltd. (n.d.) *DHT11 digital humidity and temperature sensor datasheet*. Available online (Accessed: 18 August 2026).

Used the datasheet for the 4-pin DHT11 sensor, as only the characteristics of the sensor itself were used, the project used a separate 3-pin breakout module containing the DHT11 sensor.